

Advanced Digital Design Lab
130326
Programmable Time Base Generator (Assignment 2)
Date: Tuesday, March 25, 2025

Submission:	By not more than 2 people together
Deadline:	Tuesday, April 22, 2025 (18:10) – via moodle
According to JCT's policy, if you violate any submission or deadline criteria you disqualify your assignment	

A. Programmable Time-Base Generator

The programmable time-base generator is almost identical to the hard coded time-base generator. The difference is that the counting period (i.e., number of clock cycles between every output TC pulse) is loaded from an external source into a register. The time-base generator can therefore be adopted to different frequencies or different time periods simply by loading the right value into the register.

The following figure is the block diagram of a Programmable Time-Base Generator:

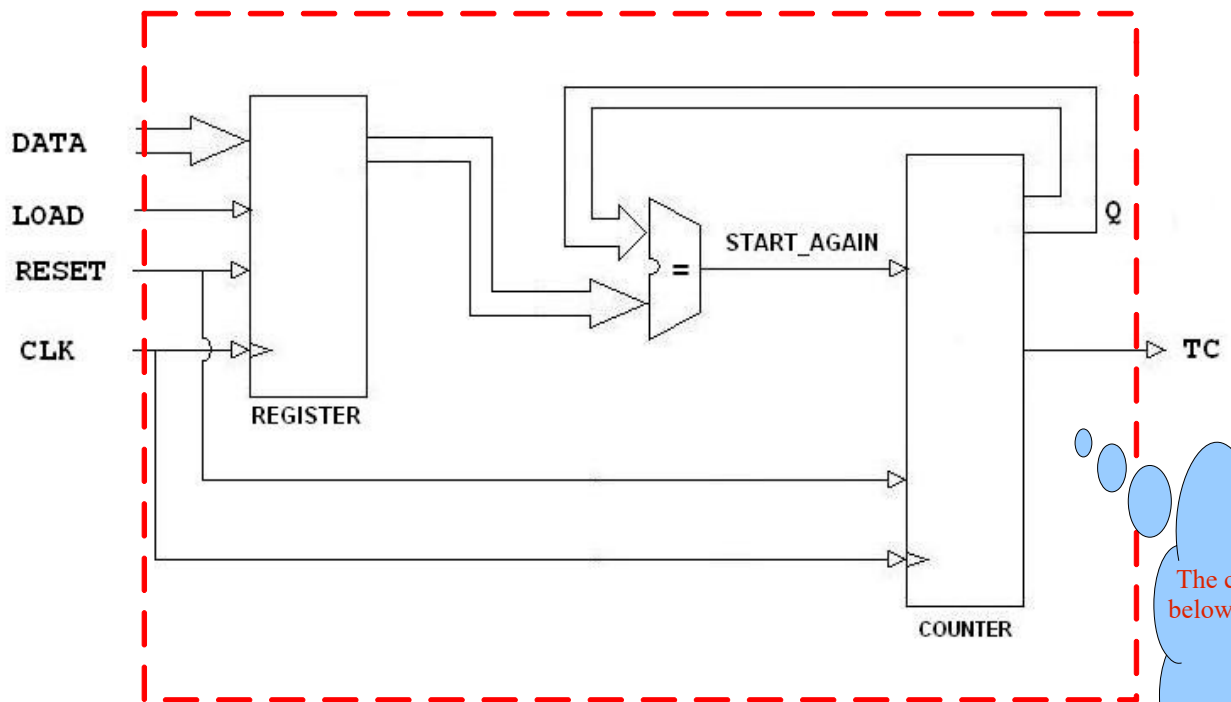


Figure 2: Programmable Time-Base Generator

The entity of the Programmable Time-Base contains only the DATA, LOAD, RESET, CLK and TC signals.

From the VHDL coding point of view it is notable to emphasize that the register portion, added now to the code, is a VHDL process that represents a register with a “clock enable”, where the LOAD signal is used for the “clock enable”.

The VHDL code below is an example for one way to design the programmable time base generator:

```

process
begin
    wait until rising_edge(CLK);
    if (RESET = '1') then
        LOADED_DATA <= (others => '0');
    else
        if (LOAD = '1') then
            LOADED_DATA <= DATA;
        end if;
    end if;
end process;

```

```

process
begin
    wait until rising_edge(CLK);
    if (RESET = '1') then
        Q <= (others => '0');
        TC <= '0';
    else
        if (Q < LOADED_DATA) then
            Q <= Q + 1;
            TC <= '0';
        else
            Q <= (others => '0');
            TC <= '1';
        end if;
    end if;
end process;

```

B. Design Requirements for the lab:

- Input clock 100MHz;
- Active high synchronous reset;
- **16** bit counter
- The loading register is **16** bits wide.

C. Lab Work

1. Simulation

Write a testbench that does the following:

1. Loads a value for 10 cycles counting
2. Waits enough time to let the counter generate 4 TC pulses
3. Loads a value for 100 cycles counting
4. Waits enough time to let the counter generate 4 TC pulses

Note: You may want to use a process in the testbench, **similar to the example given in class.**

2. Synthesis and Place & Route

2.1 Write a Xilinx XDC file for the design where:

- The LOAD signal is driven by one of the push buttons
- The DATA signal comes from the boards slide switches

2.2 Synthesize and place & route your design

- In the Pin location reports, verify that the pin locations and voltages are as expected

D. The lab report must include the following:

- VHDL source of your design
- VHDL source of your testbench
- Your XDC file
- Waveforms that show a few TC output pulses
- **Pin location report (as screenshots) from Vivado**
- Question: for a given 100MHz input clock, the 16-bit counter and the given 16 bit load register, what are the frequency ranges of the TC pulse rate?